

My research project explored the effectiveness of different supervised learning approaches on audio processing tasks. Over the course of my project, I extracted final embeddings from a self-supervised learning (SSL) model and a supervised learning (SL) model on music and speech data, looked for signs of task specialization across their intermediate layers, and compared their accuracy after training a classifier on their final embeddings. My hypothesis was that the SL model would have higher classification accuracy and exhibit less task specialization as compared to the SSL model, and my results reflect this idea but in a completely different way than I had expected.

I made several changes to my original methodology as I learned more about how DNNs work and ran into implementation constraints. First of all, my intention was to use cochleagrams as an input for the VGGish and Wav2Vec 2.0 models, but I had not realized that those two models only take waveforms as inputs. I could not find any other auditory models that a): take cochleagrams as an input, b): were pretrained on large datasets, and c): were trained on both music and speech tasks, so I decided on a compromise. Instead, I looked for models that took log-mel spectrograms as an input since they are more common, and are at least comparable to cochleagrams as time-frequency representations of audio with amplitude compression. I decided on Patchout Fast Spectrogram Transformer (PaSST) for my SL model [4] and Bootstrap Your Own Latent for Audio (BYOL-A) for my SSL model [5] as they both use log-mel spectrograms, were relatively new, and were pretrained on large datasets that included speech, music, and environmental noise.

I initially downloaded the following datasets: a small subset of the Free Music Archive, LibriVox audiobooks, and DCASE for environmental noise. Preprocessing was the most involved part of the project as I wrote several iterations of pipelines before I refined my approach. Each model required specific values of sample rate, duration, minimum and maximum pitch frequencies, and other values that required separate pipelines. Using functions from the CONAP repository [1], I generated log-mel spectrograms for all wav files I had in my data. I initially used the LibriVox dataset, but since it lacked labels that are needed to train classifiers, I instead switched to the SpeechCommands dataset and reprocessed all files [2]. I initially misunderstood PaSST as requiring log-mel spectrograms directly, but later I discovered that it takes waveforms as an input and computes log-mel spectrograms internally during feature extraction. Fortunately, the preprocessing steps I had implemented earlier were similar enough that the comparison between both models was still informative. Those steps included: normalizing amplitude, overlaying dB-adjusted SNR (signal to noise ratio) environmental noise clips over music and speech clips, and removing silence without truncating full words. This iterative learning process was quite enjoyable and helped me better understand how data preprocessing works with machine learning models. In total, I processed about 120,000 audio files across multiple iterations.

For feature extraction, I set up separate scripts for PaSST and BYOL-A to obtain both final embeddings and intermediate layer embeddings. For intermediate layers in each model, I extracted only 500 feature vectors per layer since extracting thousands of features for dozens of layers would have taken a long time. I then reduced dimensionality of each feature vector to two dimensions using PCA (PC1, PC2) and computed three separation metrics each for layer: ANOVA F/p values on PC1/PC2, silhouette scores for cluster quality, and normalized centroid

distance between music and speech centroids. These metrics later helped me look for signs of task specialization across layers.

I then extracted about ~64,000 final embeddings from each model to input into a classifier. Before training the classifier, I assigned labels to the embeddings using FMA metadata, which included genres, and the SpeechCommands file names, which included words spoken. I used an MLP classifier for both PaSST and BYOL-A, as it includes a cross-entropy loss function, which is effective for genre and word classification. I would have liked to run all 64,000 embeddings through the classifier for each model; however, I had to significantly reduce the dataset size to ~12,000 files due to hardware limitations on my local machine. Additionally, my data was skewed towards speech as I had to remove music files that had no genre label (or were classified as “unknown”).

The experiments revealed that BYOL-A exhibited task specialization across layers but had lower overall classification accuracy, while PaSST showed no task specialization and had slightly higher classification accuracy. Average classification accuracy was ~71% for BYOL-A and ~80% for PaSST. Both models performed better on word classification (~98% for BYOL-A and ~95% for PaSST) than genre classification (~68% for BYOL-A and ~61% for PaSST). Large differences in music and speech classification accuracy likely reflect dataset and label limitations, and in label quantity (FMA metadata has hundreds of labels while SpeechCommands has only a few dozen).

As for intermediate layer embeddings, BYOL-A and PaSST show substantial differences. Across all layers, PaSST consistently has silhouette scores < 0 , small centroid norms < 0.22 , and ANOVA p-values > 0.05 for PC1/PC2, indicating that music and speech embeddings do not form distinct clusters or have much internal variance between them. BYOL-A consistently has silhouette scores > 0 , ANOVA p-values < 0 for PC1, and centroid norms > 1 , which all indicate a significant separation between music and speech embeddings. Notably, these patterns were consistent across early and later layers, contrary to my expectations that task specialization would gradually increase in later layers. These results suggest that SSL models can produce task-level distinctions loosely similar to the way the auditory cortex does [3], while some SL models process audio without capturing task-level distinctions (but achieving higher accuracy).

For BYOL-A, I found it surprising that the early layers consistently showed task distinctions, so I would like to do further investigation as to why they did so. Additionally, since I did not have the opportunity to try cochleagrams in my models, I would like to design and train a custom CNN to take cochleagrams as an input and have an SSL architecture. I also recently found an interesting SSL that uses cochlear cepstograms [6], and I would like to compare it to BYOL-A and PaSST.

This research project taught me a lot about how audio works, both in how it is processed in the human ear and how it is represented in digital formats, and about the conceptual aspects of machine learning. I gained hands-on experience making preprocessing pipelines for large datasets, setting up different deep learning frameworks, working around hardware limitations, evaluating high-dimensional data using PCA, analyzing variance using ANOVA, understanding how dataset quality and labeling can strongly influence results, and about SL/SSL DNNs in general. Overall, I had a wonderful experience doing this project and will continue to do this kind of work in the future.

(Here is a link to the repository for my project: <https://github.com/mmelliem/supervised-vs-SSL>)

References

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